

Alex Baranski

Cybernetics and Artificial Intelligence PhD | Mathematics BSc
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Education

PhD in Cybernetics and Artificial Intelligence - Okinawa Institute of Science and Technology

Onna, Okinawa, Japan

2021/01 - 2026/04

- Thesis: Hyperadaptability by Behavioral Search
- Advised by Dr. Jun Tani in the Cognitive Neurorobotics Research Unit

Bachelor of Science in Mathematics - University of Chicago

Chicago, Illinois, United States

2013/09 - 2017/06

- General Honors

Professional Experience

Junior Research Fellow - Cognitive Neurorobotics Research Unit

Onna, Okinawa, Japan

2021/04 - 2026/07

- Prepared manuscript for publication on the topic of hyperadaptability.

Research Assistant - Cognitive Neurorobotics Research Unit

Onna, Okinawa, Japan

2021/01 - 2026/04

- Proposed a new theory of autonomous adaptation as search over a countably infinite dense set of behaviors.
- Used the theory to create a rapid online problem-solving algorithm based on a self-organizing graph.
- Validated the algorithm on difficult maze and dynamic control tasks against standard reinforcement learning.
- Developed differentiable graph-based spatial representations supporting fast pathfinding and adaptive behavior.
- Introduced a high-dimensional harmonic memory system for continuous spatial representation and generalization.

Computational Research Scientist - Angelo Lab

Stanford, California, United States

2018/08 - 2021/01

- Created a denoising neural network for extracting local features of low-SNR imaging data.
- Engineered MAUI, a GUI for processing multiplexed ion-beam imaging (MIBI) data.
- Developed a hierarchical anomaly-detection system for brain tissue images.
- Implemented image-to-geometry fitting algorithms for vascular imaging data, extracting quantitative descriptors such as vessel thickness and curvature.
- Created a statistical model of signal heterogeneity in MIBI data to support quantitative spatial analysis.
- Made a mathematical model of field-intensity artifacts in MIBI data.
- Built a variational autoencoder for quantitative analysis of single cell morphology.

Software Engineer - 4th Law

San Jose, California, United States

2017/07 - 2018/04

- Developed a fractal database model for efficient storage of multi-scale geospatio-temporal data.
- Engineered a modular control platform for fleets of autonomous robotic systems.
- Worked with control engineer to design adaptive control systems.
- Wrote technical recommendations for the CTO and head engineer.

Undergraduate Researcher - MacLean Lab

Chicago, Illinois, United States

2015/05 - 2015/08

- Created model of transient excitation in anesthetized mouse brain using a network of adaptive exponential integrate-and-fire neurons.

Student Technician - Freedman Lab

Chicago, Illinois, United States

2014/05 - 2014/08

- Wrote real-time acquisition software for neural and eye-tracking data from Rhesus macaques.

Skills

- **Research:** Project Management, Creative Problem-Finding, Creative Problem-Solving, Mathematical Thinking
- **Programming:** Python, PyTorch, Java, MATLAB, JavaScript, TypeScript, Bash, basic C++ and C
- **Artificial Intelligence:** Deep Learning, Generative Modeling, Genetic Algorithms, Reinforcement Learning, Planning, Computer Vision, Architecture Design, Sampling Methods
- **Mathematics:** Abstract Algebra, Function Approximation, Dynamical Systems, Numerical Techniques, Harmonic Methods, Implicit and Parametric Geometric Representations, Graph Theory, Combinatorics, Calculus
- **Computer Science:** Complexity Analysis, Algorithm Design, Data Structures, Software Engineering, Geometric Modeling, Information Theory, Automata Theory
- **Neuroscience:** Cognitive, Evolutionary, Developmental, Systems, Cellular
- **Tools:** MuJoCo, SymPy, Mathematica, Slurm, Git, Inkscape, LaTeX, Blender

Publications (First-Author)

- **Baranski, A.**, Jun, T. *Life, uh, Finds a Way: Hyperadaptability by Behavioral Search*. arXiv, 2025
- **Baranski, A.**, Froese, T. *Efficient Spike Timing Dependent Plasticity Rule for Complex-Valued Neurons*. ALIFE 2021: The 2021 Conference on Artificial Life
- **Baranski, A.**, Milo, I., Greenbaum, S., Oliveria, J., Mrdjen, D., Angelo, M., Keren, L. *MAUI (MBI Analysis User Interface)—An image processing pipeline for Multiplexed Mass Based Imaging*. PLoS Computational Biology, 2021

Publications (Co-authored)

- Risom, T., Rivero, B., Liu, C., **Baranski, A.**, Strand, S., Greenwald, N., McCaffrey, E., Varma, S., Keren, L., Srivastava, S., Zhu, C., Vennam, S., Hwang, S., Colditz, G., Bendall, S., West, R., Angelo, M. *Mapping the tumor and microenvironmental evolution underlying DCIS progression through multiplexed ion beam imaging*. Cancer Research, 2020
- Risom, T., Glass, D., Averbukh, I., Liu, C., **Baranski, A.**, Kagel, A., McCaffrey, E., Greenwald, N., Gutiérrez, B., Strand, S., Varma, S., Kong, A., Keren, L., Srivastava, S., Zhu, C., Khair, Z., Veis, D., Deschryver, K., Vennam, S., Maley, C., Hwang, E., Marks, J., Bendall, S., Colditz, G., West, R., Angelo, M. *Transition to invasive breast cancer is associated with progressive changes in the structure and composition of tumor stroma*. Cell, 2022
- Vijayaragavan, K., Cannon, B., Tebaykin, D., Bossé, M., **Baranski, A.**, Oliveria, J., Mrdjen, D., Corces, M., McCaffrey, E., Greenwald, N., Sigal, Y., Khair, Z., Bruce, T., Rajaraman, A., Bukhari, S., Montine, K., Angelo, R., Montine, T., Bendall, S. *Single-cell Spatial Proteomic Imaging for Human Neuropathology*. bioRxiv, 2022
- McCaffrey, E., Donato, M., Keren, L., Chen, Z., Delmastro, A., Fitzpatrick, M., Gupta, S., Greenwald, N., **Baranski, A.**, Graf, W., Kumar, R., Bosse, M., Fullaway, C., Ramdial, P., Forgó, E., Jovic, V., Valen, D., Mehra, S., Khader, S., Bendall, S., Rijn, M., Kalman, D., Kaushal, D., Hunter, R., Banaei, N., Steyn, A., Khatri, P., Angelo, M. *The immunoregulatory landscape of human tuberculosis granulomas*. Nature Immunology, 2022
- Ferrian, S., Liu, C., McCaffrey, E., Kumar, R., Nowicki, T., Dawson, D., **Baranski, A.**, Glaspy, J., Ribas, A., Bendall, S., Angelo, M. *Multiplexed imaging reveals an IFN- γ -driven inflammatory state in nivolumab-associated gastritis*. Cell Reports Medicine, 2021
- Hartmann, F., Mrdjen, D., McCaffrey, E., Glass, D., Greenwald, N., Bharadwaj, A., Khair, Z., Verberk, S., **Baranski, A.**, Baskar, R., Graf, W., Valen, D., Bossche, J., Angelo, M., Bendall, S. *Single-cell metabolic profiling of human cytotoxic T cells*. Nature Biotechnology, 2021

Presentations

- **A Baranski**. *Generating Adaptive Behavior by a Self-Mutating Search Process*. Presented at Neuroscience 2024.
- **A Baranski**, J Tani. *Behavior-Model Coevolution with Working Memory*. Presented at the International Workshop on Cognitive Neurorobotics 2024.
- **A Baranski**, J Tani. *Quasiparametric Networks with Fast Weight Memory for Spatial Search*. Presented at the International Symposium on Artificial Intelligence and Brain Science 2022.
- **A Baranski**, D Mjurdén, JP Oliveria, K Vijayaragavan, TJ Montine, SC Bendall and M Angelo. *Automatic pattern differentiation in multiplexed imaging data*. Presented at BioImage Informatics 2019.
- B Cannon, **A Baranski**, K Vijayaragavan, D Mrdjen, J Oliveria, and S Bendall. *Capturing human brain tissue phenotypes with pixel-connected object identification in multiplexed ion beam imaging data*. Presented at BioImage Informatics 2019.